

4,6,10,Slack

Program 3: Create a maven projects with all dependencies required for the application in CI/CD pipeline.

MAVEN PROJECT:

Step1:

Open Eclipse

File -> new-> maven project ->next

Choose maven-archetype-quickstart -> version 1.1 or 1.4

Group id nd artifact id anything, Artifact nd package name should be same

In chrome, search maven repository

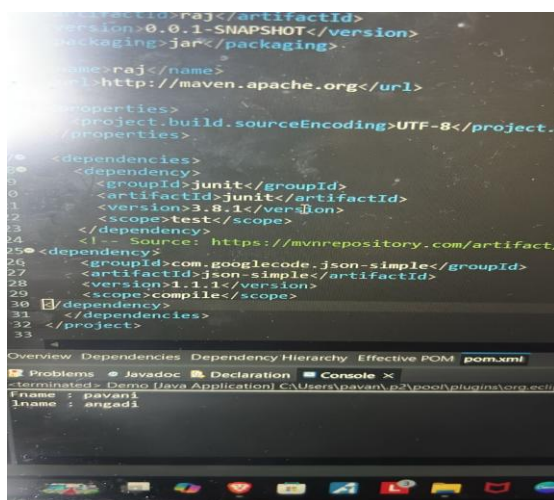
Search json simple

1st json.simple

Click the one which has more usages. 1.1.1

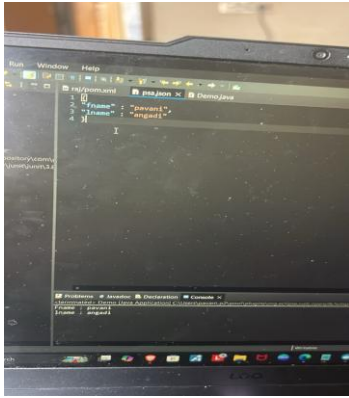
In the maven u will have, there will be dependency code

pom.xml paste the dependency code



Right click on package -> new -> file -> filename: any.json

Code to put in .json file:



Right click on src/test/java

New-> class -> Demo.java

Code to put in Demo.java:

```
import java.io.FileReader;
```

```
import org.json.simple.JSONObject;
```

```
import org.json.simple.parser.JSONParser;
```

```
public class Demo {
```

```
    public static void main(String[] args) {
```

```
        JSONParser p = new JSONParser();
```

```
        try {
```

```
            Object obj = p.parse(new FileReader("san.json"));
```

```
            JSONObject jo = (JSONObject) obj;
```

```
            String fname = (String) jo.get("fname");
```

```
            String lname = (String) jo.get("lname");
```

```
            System.out.println("Fname : " + fname);
```

```
        System.out.println("Iname : " + Iname);

    }

    catch (Exception e) {
        e.printStackTrace();
    }
}
}
```

Save and run

1st Program:

Create a folder nd open in vs code

Create Dockerfile

Content in Dockerfile:

FROM eclipse-temurin:21-jdk

WORKDIR /app

COPY . /app

RUN javac prime.java

CMD ["java","prime"]

Put the java file name instead of prime

Create .java file

Write any apln code

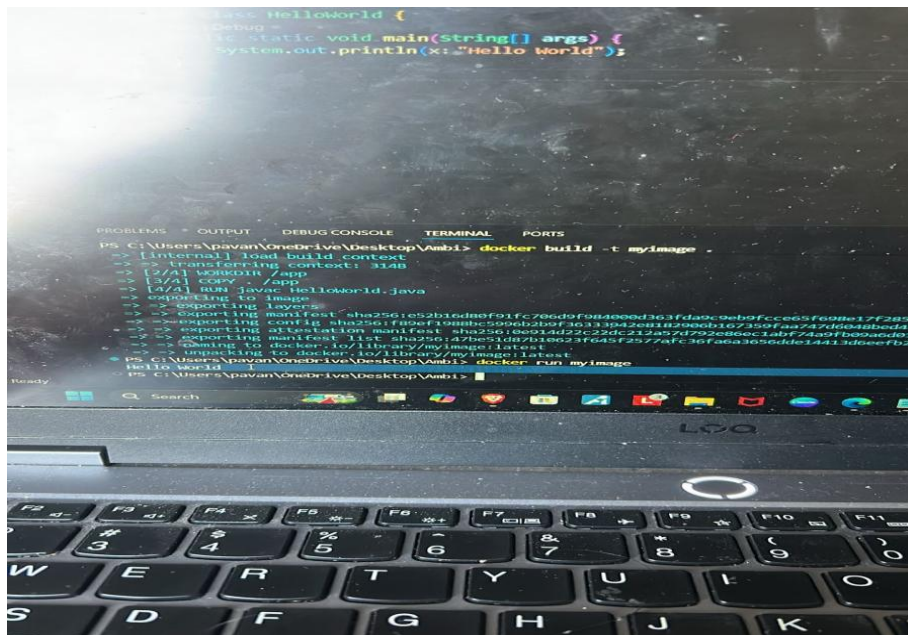
Save

Run these 2 commands in terminal:

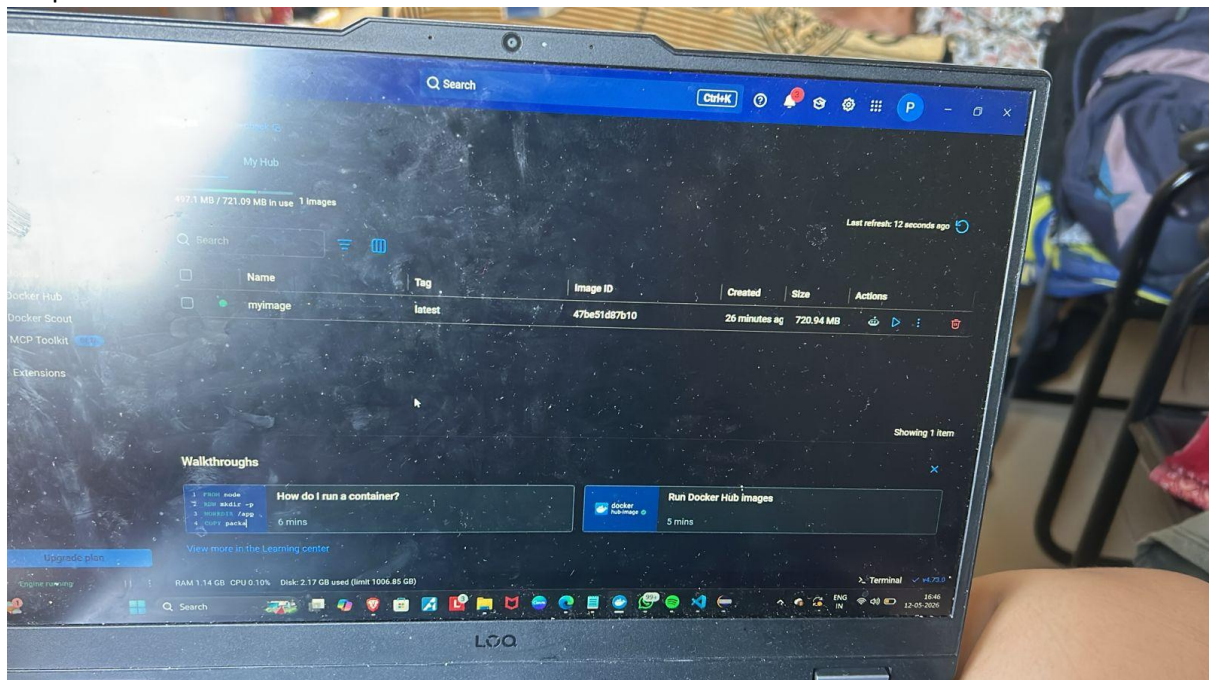
docker build -t h1 .

//where -t h1 is image name

docker run image-name



Output :



Docker hub should be running-> engine should be running

Image should be in docker desktop

2nd Program: Create and configure Jenkins files for workflow and build of an application and push the image

Create a new repo in github

Create a Dockerfile with the content:

```
FROM eclipse-temurin:21-jdk
```

```
WORKDIR /app
```

```
COPY . /app
```

```
RUN javac prime.java
```

```
CMD ["java","prime"]
```

Create a java appIn file (say Hello.java)

Create a Jenkinsfile with the content:

```
pipeline {
```

```
    agent any
```

```
    environment {
```

```
        DOCKERHUB_CREDENTIALS = 'Docker-credentials'
```

```
        IMAGE_NAME = 'pannu27/new_docker_image'
```

```
    }
```

```
    stages {
```

```
        stage('Build Java Application') {
```

```
            steps {
```

```
                bat 'javac prime.java'
```

```
            }
```

```
        }
```

```
        stage('Run Java Program') {
```

```
            steps {
```

```
                bat 'java prime'
```

```
            }
```

```
        }
```

```

stage('Build Docker Image') {
    steps {
        bat 'docker build -t %IMAGE_NAME%:latest .'
    }
}

stage('Login to DockerHub') {
    steps {
        withCredentials([usernamePassword(
            credentialsId: 'Docker-credentials',
            usernameVariable: 'USER',
            passwordVariable: 'PASS')]) {

            bat 'echo %PASS% | docker login -u %USER% --password-stdin'

        }
    }
}

stage('Push Docker Image') {
    steps {
        bat 'docker push %IMAGE_NAME%:latest'
    }
}
}

```

Go to Jenkins -> manage credentials -> add credential -> username p.w

Give username and p.w of dockerhub and click Create

Jenkins->new item -> pipeline project -> ok

Select pipelinescript from scm

Scm -> select git

Add git repo link

Add credential

Branch -> main

Apply and save

Click on build now

It should show green check with all the stages being completed

Image will be showed in docker hub

4th program: slack

Open Jenkins

Open slack.com

It should show welcome to lab practice

Create new channel

More -> tools-> apps -> search Jenkins -> open app -> configure -> add to slack -> select channel nd integrate -> go to step3 -> copy the token

Go to Jenkins

Available plugins -> select slack notification and global slack notifier and install them.

No need to restart Jenkins

Go to settings -> manage Jenkins ->system ->slack

Workspace : csections-workspace

Credentials: Add -> global -> secret text (the token u copied) ->id:anything and then create

Click drop down nd select the credential

Test connection -> should show success

Apply and save

Click on new item

Select free style project -> ok

Build steps: Add build steps -> execute windows batch command -> java -version

Post build actions:

Add post build action -> slack notifications

Select:

Notify build start

Notify success

Notify abort

And then click on build now